Donut iso variants: inner-R truth-purity scan

All-calo 120 MeV donut with $R_{\text{outer}} = 0.4$ (cross-check)

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Executive summary

Two new reco-isolation definitions were added as cross-check variants (`syst_type=None`, `syst_role=None`):

- **donutFull** (`use_topo_iso=4`): `cluster_iso_04` − `cluster_iso_{005,0075,02}`. All-calo 120 MeV tower sum (EMCal+HCalIn+HCalOut), outer $R = 0.4$, inner $R \in \{0.05, 0.075, 0.20\}$. Cluster ET is subtracted as a scalar in both terms; the two subtractions cancel in the difference.
- **donutExcl** (`use_topo_iso=5`): `cluster_iso_excl_04` − `cluster_iso_excl_{005,0075,01,02}`. Same three-calo 120 MeV sum, but EMCal towers owned by the cluster are skipped at tower granularity rather than subtracted as a scalar. Inner $R \in \{0.05, 0.075, 0.10, 0.20\}$.

Switch added in `RecoEffCalculator_TTreeReader.C:2042-2065` and mirrored in five companion macros. The `donutFull_01` variant is deliberately skipped because `cluster_iso_01` (raw family, $R = 0.10$) does not exist in the current slimtree; logged in `reports/anatreemaker_reprocess_TODO.md` along with the EMCal-only inner-ring branches needed for a pure-EMCal donut.

Two-pass workflow

Per-variant iso cut was derived at 80 % target efficiency:

1. **Pass 1**: run the full `RecoEffCalculator_TTreeReader + MergeSim + CalculatePhotonYield` pipeline with placeholder loose cut ($b = 100$, $s = 0$, i.e. no iso cut). Produces `MC_efficiency_bdt_donut*.r` with the unbiased 2D distribution `h_singal_reco_isoET_0`.
2. **Cut derivation**: parameterized `FindETCut.C(var_type, target_eff)` fits a linear $E_T^{\text{iso,cut}} = b + s\,p_T$ at 80 % efficiency; `derive_iso_cuts.py` drives all 7 variants and writes `derived_iso_cuts.yaml`.
3. **Pass 2**: regenerate configs with derived (b, s), re-run pipeline on HTCondor cluster 2701654.

Table 1: Derived 80 % iso cuts (isoET $< b + s\,p_T$).

Variant	b [GeV]	s
donutFull_005	0.1089	0.0224
donutFull_0075	0.2642	0.0061
donutFull_02	0.2305	−0.0012
donutExcl_005	0.2457	0.0111
donutExcl_0075	0.2774	0.0053
donutExcl_01	0.2921	0.0017
donutExcl_02	0.2305	−0.0012

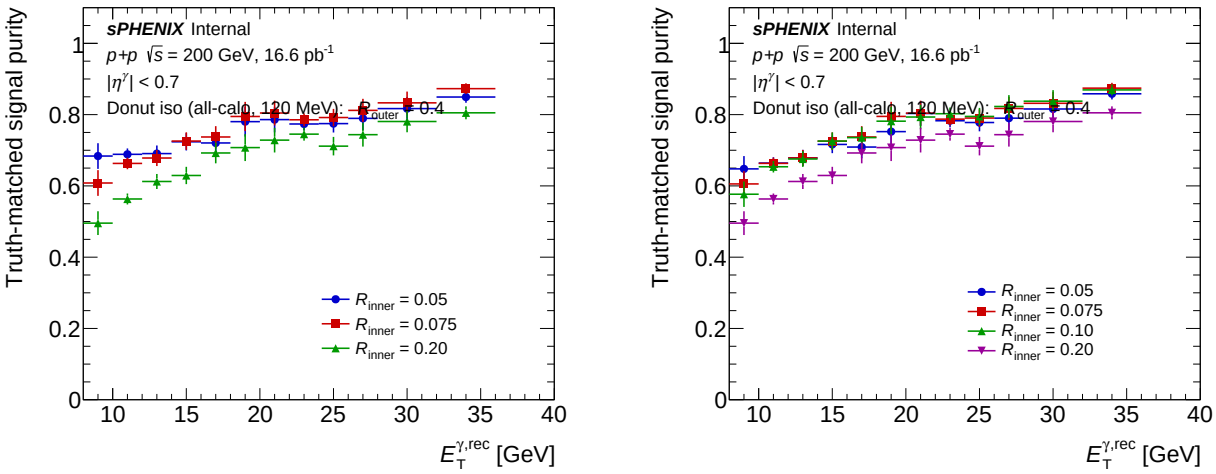
Note: `donutFull_02` and `donutExcl_02` land on identical fit bins. At $R_{\text{inner}} = 0.2$ the cluster footprint is fully contained within the excluded EMCal tower set, so ownership-exclusion and scalar-E_T subtraction yield the same annulus integral.

Truth-matched signal purity (g_purity_truth)

Purity is read from `Photon_final_bdt_<var>_mc.root` (MC-as-data, truth labels populated). The non-`_mc` file returns zero because `_signal` histograms are absent from data.

Table 2: Truth-matched photon purity at three representative p_T bins. Nominal topo04 iso included for reference.

Variant	$y(9\,\text{GeV})$	$y(21\,\text{GeV})$	$y(34\,\text{GeV})$
nom (reference)	0.581	0.743	0.797
donutFull_005	0.684	0.786	0.849
donutFull_0075	0.608	0.805	0.873
donutFull_02	0.496	0.728	0.805
donutExcl_005	0.648	0.803	0.858
donutExcl_0075	0.606	0.804	0.874
donutExcl_01	0.577	0.793	0.869
donutExcl_02	0.496	0.728	0.805



(a) donutFull: $R_{\text{inner}} \in \{0.05, 0.075, 0.20\}$.

(b) donutExcl: $R_{\text{inner}} \in \{0.05, 0.075, 0.10, 0.20\}$.

Figure 1: Truth-matched signal purity vs. cluster p_T for the donut-iso variants. Curves rise from ~ 0.5–0.7 at 8 GeV to ~ 0.80–0.87 at 34 GeV. Tighter inner R (smaller value) yields higher purity at low p_T; variants converge at high p_T.

Physics observations

- All donut variants with $R_{\text{inner}} \leq 0.10$ reach higher purity than the nominal topo04 iso at all p_T bins.
- Excl family at $R_{\text{inner}} = 0.05$ sits between nominal and Full_005 at low p_T, converging at high p_T. The ownership-based exclusion removes slightly less annulus-nearby energy than scalar $-E_T$ does, translating into a marginally looser effective iso.

- `donutFull_02` = `donutExcl_02` identically to 4 decimals across all 12 p_T bins, confirming the same derived iso cut and truth purity.
- Across the Excl family at 9 GeV, purity decreases monotonically $0.648 \to 0.606 \to 0.577 \to 0.496$ as R_{inner} grows $0.05 \to 0.075 \to 0.10 \to 0.20$. Physical: larger R_{inner} means a smaller annulus and a looser absolute iso cut, admitting more background.

Caveats and follow-ups

- **Outer cone includes HCal.** The donut is not pure EMCal. A pure-EMCal donut requires new inner-ring branches `cluster_iso_{005,0075,01,02}_emcal` at matching threshold, logged in `reports/anatreemaker_reprocess_TODO.md`.
- `donutFull_01` **skipped** pending upstream `cluster_iso_01` branch (all-calo, 120 MeV, $R < 0.1$).
- **Degraded mirrors** (`Cluster_rbr.C`, `EtaMigrationStudy.C`, `compare_energy_response.C`) silently use the full outer cone without inner subtraction for cases 4/5 because they do not read `iso_emcalinnerr`. These are diagnostic utilities only; do not run donut variants through them expecting donut semantics.
- **MC iso scale/shift** are set to 1.0/0.0 for all donut variants because the nominal 1.2/0.1 scale/shift was tuned against the full-cone topo04 iso. Retuning would be needed if these became systematic (as opposed to cross-check) variants.

Reviewer pass matrix

Artifact	Physics	Cosmetics	Re-derivation	Fix-validator
`use_topo_iso=4,5` switch	PASS (2 WARN util)	N/A	N/A	PASS
donutFull scan PDF	N/A	PASS (2 WARN)	PASS	N/A
donutExcl scan PDF	N/A	PASS (2 WARN)	PASS	N/A
80 % iso cuts	N/A	N/A	PASS	PASS
`g_purity_truth` (7 variants)	N/A	N/A	PASS	PASS